

作业 (4月15日)

1  
3 (要求写出带佩亚诺余项的泰勒公式)

4  
5 (1)

补充作业

求  $z = x^2 + y^2$  在  $d(x, y) | x^2 + y^2 \leq 1$  上的最大和最小值.

作业 (4月17日)

p121 习题8-6

5

6 (2) (3)

11

12

13

p128-129 第八章综合题  
11

$$4. (2) \quad z = x^4 + y^4 - x^2 - 2xy - y^2$$

$$\begin{cases} \frac{\partial z}{\partial x} = 4x^3 - 2x - 2y = 0 & ① \\ \frac{\partial z}{\partial y} = 4y^3 - 2y - 2x = 0 & ② \end{cases}$$

$$① - ② \quad x^3 = y^3 \Rightarrow x = y$$

$$\text{代入 } ① \quad x^3 = x \quad x = \pm 1, 0$$

$$\frac{\partial^2 z}{\partial x^2} = 12x^2 - 2 \quad \frac{\partial^2 z}{\partial x \partial y} = -2 \quad \frac{\partial^2 z}{\partial y^2} = 12y^2 - 2$$

$$① \quad (1, 1) \text{ 与 } (-1, -1) \quad A=10 \quad B=-2 \quad C=10 \quad B^2 - AC < 0, \quad AC > 0, \quad \text{极值点} \quad z = -2$$

$$② \quad (0, 0) \quad A=B=C=-2 \quad B^2 - AC = 0 \quad \text{定理 8.13 无法判断}$$

$$\text{回归原式: } z = x^4 + y^4 - (x+y)^2 \quad z_{(0,0)} = 0$$

$$x=y \rightarrow 0 \quad z = 2x^4 - 4x^2 < 0$$

$$x=-y \rightarrow 0 \quad z = 2x^4 > 0$$

}  $\Rightarrow (0, 0)$  不为极值点 (鞍点)

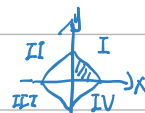
6(2)  $u = x^2 - xy + y^2 \quad |x| + |y| \leq 1$

$$\begin{cases} \frac{\partial u}{\partial x} = 2x - y = 0 \\ \frac{\partial u}{\partial y} = -x + 2y = 0 \end{cases} \Rightarrow \text{驻点 } (0,0) \Rightarrow \begin{cases} \frac{\partial^2 u}{\partial x^2} = 2 \\ \frac{\partial^2 u}{\partial x \partial y} = -1 \\ \frac{\partial^2 u}{\partial y^2} = 2 \end{cases} \quad B^2 - AC < 0, A, C > 0, (0,0) \text{ 为极小值点}$$

$\boxed{u_{(0,0)} = 0}$

极值只能在驻点上取得，但最值不一定，故还要考虑边界

分象限考虑，现考察第一象限的边界  $x+y=1, x, y \geq 0$



条件极值: Lagrange 乘数法:  $F(x, y, \lambda) = x^2 - xy + y^2 - \lambda(x+y-1)$

$$\begin{cases} F'_x = 2x - y - \lambda = 0 \quad (1) \\ F'_y = -x + 2y - \lambda = 0 \quad (2) \\ F'_\lambda = x + y - 1 = 0 \quad (3) \end{cases} \Rightarrow \begin{aligned} (1)+(2): 2\lambda = x+y-1 \quad \lambda = \frac{1}{2} \\ x=y=\lambda = \frac{1}{2} \end{aligned}$$

$\boxed{u(\frac{1}{2}, \frac{1}{2}) = \frac{1}{4}}$  为  $u = x^2 - xy + y^2$  在条件  $x+y=1 (x, y \geq 0)$  上的极值 (小)

考虑条件  $x+y=1 (x, y \geq 0)$  的边界  $(0,1) (1,0) \quad u(0,1) = u(1,0) = 1$

其他象限同理，综上  $u_{\min} = 0 \quad u_{\max} = 1$

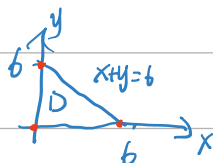
① 最值一定要考虑边界

② 二维区域  $D$  的边界  $\partial D$  为“边” (一维);  $\partial D$  也有自己的边界“顶点” (0维)

③ 边界上的条件极值只能算边界上的极值，不是整个区域上的极值

第八章综合题 11, 同理

11.  $z = f(x, y) = x^2 y (4 - x - y)$  区域  $D$



求最值需考虑

① 区域  $D$  内部的驻点 只有这个才有可能叫闭区域  $D$  上的极值  $f(2,1) = 4$  极大值, 最大值

② 三条边界上的条件极值点  $f(4,2) = -16$  最小值

③ 条件极值的边界 (三个顶点)

## 微积分习题课材料

1 求由方程  $x^2 + y^2 + z^2 - 2x + 2y - 4z - 10 = 0$

确定的函数  $z = f(x, y)$  的极值.

2 求函数  $u = x + y + z$  在沿球面  $x^2 + y^2 + z^2 = 1$  上的点  $P_0(x_0, y_0, z_0)$  处的外法线方向上

的导数  $\frac{\partial u}{\partial n}$ , 问球面上哪点使  $\frac{\partial u}{\partial n}$

最大, 最小, 等于零?

3 若函数  $f(x, y)$  对任意的正实数  $t$  满足关系

$f(tx, ty) = t^n f(x, y)$ , 则称  $f(x, y)$  为  $n$  次齐次函数,

设  $f(x, y)$  可微, 试证明  $f(x, y)$  为  $n$  次齐次函数的充要条件是  $xf_x(x, y) + yf_y(x, y) = nf(x, y)$ .

4 设  $u = u(x, y)$  是  $C^{(2)}$  类函数, 作极坐标变换

$x = r \cos \theta, y = r \sin \theta$ , 试把

$$W_1 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 \text{ 和 } W_2 = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

转化为极坐标系下的形式

5 给定曲面  $F\left(\frac{x-a}{z-c}, \frac{y-b}{z-c}\right) = 0$  ( $a, b, c$  为常数),

或由它确定的曲面  $z = z(x, y)$ , 这里

$F$  是  $C^{(1)}$  类函数,  $z(x, y)$  是  $C^{(2)}$  类函数,

试证明:

(1) 原曲面的切平面通过一个定点;

(2) 函数  $z = z(x, y)$  满足方程

$$\frac{\partial^2 z}{\partial x^2} \cdot \frac{\partial^2 z}{\partial y^2} - \left(\frac{\partial^2 z}{\partial x \partial y}\right)^2 = 0.$$

1.

法1. 规则法

$$F(x, y, z) = x^2 + y^2 + z^2 - 2x + 2y - 4z - 10$$

$$z_x = \frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{x-1}{z-2} \quad z_{xx} = \frac{\partial}{\partial x} \left( \frac{x-1}{z-2} \right)$$

或者  $F(x, y, z) = 0$  两边对  $x$  求两次导

$$\text{求出 } z_x, z_y, z_{xx}, z_{xy}, z_{yy} \quad [z_x = z_y = 0 \Leftrightarrow B^2 - AC > 0 \text{ 判断极值}]$$

略

另法, 画立方, 几何意义:  $(x-1)^2 + (y-1)^2 + (z-2)^2 = 16$  球

$$z-2 \in [-4, 4]$$

$$2. \quad x = r \cos \alpha \cos \beta \quad y = r \cos \alpha \sin \beta \quad z = r \sin \alpha \quad \alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

极坐标换元

$$\beta \in (-\pi, \pi)$$

$$\text{球面 } x^2 + y^2 + z^2 = 1 \Leftrightarrow r = 1$$

$$u = r \cos \alpha \cos \beta + r \cos \alpha \sin \beta + r \sin \alpha$$

$$\boxed{\dots} \frac{\partial u}{\partial r} = \frac{\partial u}{\partial r} \Big|_{r=1} = \cos \alpha \cos \beta + \cos \alpha \sin \beta + \sin \alpha = x_0 + y_0 + z_0 \leq 3 \sqrt{\frac{x_0^2 + y_0^2 + z_0^2}{3}} = \sqrt{3}$$

$$\frac{\partial u}{\partial r} \in [-\sqrt{3}, \sqrt{3}]$$

$$\sqrt{3}: (x_0, y_0, z_0) = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$$

$$-\sqrt{3}: (x_0, y_0, z_0) = \left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$$

0: 平面  $x_0 + y_0 + z_0 = 0$  与球面  $x_0^2 + y_0^2 + z_0^2 = 1$  的交线

3.  $\Rightarrow f(tx, ty) = t^n f(x, y)$  对  $\forall t$  成立,

两边同时对  $t$  求导:  $LHS = x f_x(tx, ty) + y f_y(tx, ty) = n t^{n-1} f(x, y) = RHS$

令  $t=1$ , 即得  $x f_x(x, y) + y f_y(x, y) = n f(x, y)$

令  $F(t) = f(tx, ty) - t^n f(x, y)$  我们证明  $\forall t, F(t) = 0$

而  $F(1) = 0, F'(t) = x f_x(tx, ty) + y f_y(tx, ty) - n t^{n-1} f(x, y)$

条件是  $x f_x(x, y) + y f_y(x, y) = n f(x, y) \quad F'(1) = F(1) = 0$

找出  $F$  和  $F'$  的关系构造微分方程求解  $F$

$$\Rightarrow t F'(t) = t x f_x(tx, ty) + t y f_y(tx, ty) - n t^n f(x, y) = n t f(tx, ty) - n t^n f(x, y) = n F(t)$$

$$t \cdot \frac{dF}{dt} = nF \Rightarrow F = Ct^n$$

$$F(1) = F(1) = 0 \text{ 代入得 } C = 0$$

$$\text{即 } F(t) = 0 \text{ ④}$$

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4 设  $u = u(x, y)$  是  $C^{(2)}$  类函数, 作极坐标变换  $x = r \cos \theta$ ,  $y = r \sin \theta$ , 试把

$$W_1 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 \text{ 和 } W_2 = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

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$$r = \sqrt{x^2 + y^2} \quad \theta = \arctan \frac{y}{x}$$

$$\textcircled{1} \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial x}$$

$$\frac{\partial r}{\partial x} = \frac{1}{\sqrt{x^2 + y^2}} \cdot 2x = \frac{x}{r} = \cos \theta$$

$$\begin{cases} y = \tan \theta & \frac{dy}{d\theta} = \tan^2 \theta + 1 = y^2 + 1 \\ x = \arctan y & \frac{dx}{dy} = \frac{1}{y^2 + 1} \end{cases}$$

$$\frac{\partial \theta}{\partial x} = \frac{1}{y^2 + 1} \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{r^2} = -\frac{\sin \theta}{r}$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{x}{r} \frac{\partial u}{\partial r} - \frac{y}{r^2} \frac{\partial u}{\partial \theta} = \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \textcircled{1}$$

$$\text{同理 } \frac{\partial u}{\partial y} = \frac{y}{r} \frac{\partial u}{\partial r} + \frac{x}{r^2} \frac{\partial u}{\partial \theta} = \sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \textcircled{2}$$

$$W_1 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta}\right)^2$$



$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \textcircled{1} = \frac{\partial}{\partial r} \textcircled{1} \frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta} \textcircled{1} \frac{\partial \theta}{\partial x} = \cos \theta \frac{\partial}{\partial r} \textcircled{1} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \textcircled{1}$$

$$\frac{\partial}{\partial x} \left( \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) = \cos \theta \frac{\partial^2 u}{\partial r^2} + \frac{\sin \theta}{r^2} \frac{\partial u}{\partial \theta} - \frac{\sin \theta}{r} \frac{\partial^2 u}{\partial \theta \partial r}$$

$$\frac{\partial}{\partial \theta} \left( \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) = -\sin \theta \frac{\partial u}{\partial r} + \cos \theta \frac{\partial^2 u}{\partial r \partial \theta} - \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} - \frac{\sin \theta}{r} \frac{\partial^2 u}{\partial \theta^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \cos^2 \theta \frac{\partial^2 u}{\partial r^2} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\sin^2 \theta}{r} \frac{\partial u}{\partial r}$$

$$\text{同理 } \frac{\partial^2 u}{\partial y^2} = \sin^2 \theta \frac{\partial^2 u}{\partial r^2} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\cos^2 \theta}{r} \frac{\partial u}{\partial r}$$

$$\Rightarrow W_2 = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{1}{r} \frac{\partial u}{\partial r}$$



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5. 盲猜  $(a, b, c)$

观察  $F$  的形式

隐函数标准形式:  $G(x, y, z) = 0 \quad [G(x, y, z) = F\left(\frac{x-a}{z-c}, \frac{y-b}{z-c}\right)]$

$$\text{其中 } G_x = F' \cdot \frac{1}{z-c} \quad G_y = F' \cdot \frac{1}{z-c} \quad G_z = -\frac{x-a}{(z-c)^2} F' - \frac{y-b}{(z-c)^2} F'$$

曲面法:  $\vec{n} = (G_x, G_y, G_z)$

任取曲面上一点  $M_0(x_0, y_0, z_0)$ , 该点法向量  $\vec{n}_0 = (G_{x_0}, G_{y_0}, G_{z_0})$

$$\text{切平面 } \frac{F'_1}{z_0-c} \cdot (x-x_0) + \frac{F'_1}{z_0-c} (y-y_0) - \frac{F'_1}{(z_0-c)^2} (x_0-a)(z-z_0) - \frac{F'_2}{(z_0-c)^2} (y_0-b)(z-z_0) = 0$$

$(a, b, c)$  代入验证即可

或者强行化简

$$\Rightarrow \frac{F'_1}{z_0-c} [x-x_0 - \frac{z-z_0}{z_0-c} (x_0-a)] + \frac{F'_2}{z_0-c} [y-y_0 - \frac{z-z_0}{z_0-c} (y_0-b)] = 0$$

$$\Rightarrow \frac{F'_1}{z_0-c} [(x-a) - \frac{x_0-a}{z_0-c} (z-c)] + \frac{F'_2}{z_0-c} [(y-b) - \frac{y_0-b}{z_0-c} (z-c)] = 0$$

另法: 几何关系:

$$(x_0-a, y_0-b, z_0-c) \cdot (G_{x_0}, G_{y_0}, G_{z_0}) = 0$$

$A(a, b, c) \quad M_0(x_0, y_0, z_0)$  也就是说  $\vec{AM}_0 \perp \vec{n}_0$ ,  $A$  在  $M_0$  的切面上, 而  $M_0$  是任意的

$$(2) \quad \frac{\partial z}{\partial x} = -\frac{G_x}{G_z} = \frac{F'_1}{\frac{x-a}{z-c} F' + \frac{y-b}{z-c} F'_2}$$

$$\frac{\partial z}{\partial y} = \frac{F'_2}{\frac{x-a}{z-c} F' + \frac{y-b}{z-c} F'_2} \quad (\text{不更算})$$

另: 利用切平面

$$G_x(x-a) + G_y(y-b) + G_z(z-c) = 0$$

$$\Rightarrow z-c = z_x(x-a) + z_y(y-b) \quad \text{两边对 } x, y \text{ 求偏导}$$

$$\begin{cases} z_x = z_{xx}(x-a) + z_x + z_{yx}(y-b) \\ z_y = z_{xy}(x-a) + z_y + z_{yy}(y-b) \end{cases} \Rightarrow \begin{cases} (x-a)z_{xx} = -(y-b)z_{yx} \quad (1) \\ (y-b)z_{yy} = -(x-a)z_{yx} \quad (2) \end{cases}$$

$$(1) \times (2) \quad (x-a)(y-b) \neq 0 \quad z_{xx}z_{yy} = (z_{yx})^2$$